2.3 Ordering of Integers

We have already seen the ordering of numbers in the set of natural and whole numbers. The ordering is possible for integers also.

2.3.1 Predecessor and Successor of an Integer

Recall that for a given number its predecessor is one less than it and its successor is one more than it. This applies for integers also.

Example 5 Find the predecessor and successor of i) 0 and ii) −8 on a number line.

Solution

Place the given numbers on the number line then move one unit to their right and left to get the successor and the predecessor respectively.



We can see that the successor of 0 is +1 and the predecessor of 0 is −1 and the successor of −8 is −7 and the predecessor of −8 is −9.

Note

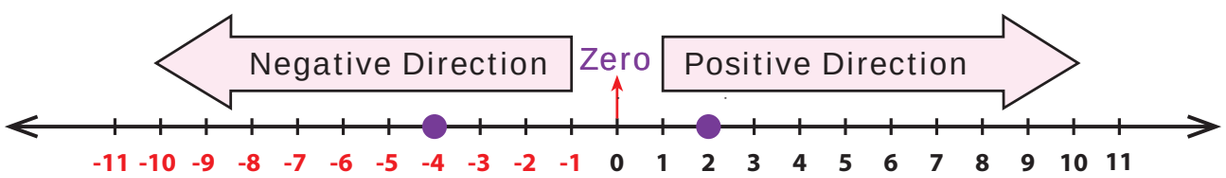
- Every positive integer is greater than each of the negative integers. Example: 3 > −5
- 0 is less than every positive integer but greater than every negative integer. Example: 0<2 but 0> −2

2.3.2 Comparing Integers

Ordering of integers is to compare them. It is very easy to compare and order integers by using a number line.

When we move towards the right of a number on the number line, the numbers become larger. On the other hand, when we move towards the left of a number on the number line, the numbers become smaller.

We know that 4 < 6 , 8 > 5 and so on. Now let us consider two integers say −4 and 2. Mark them on the number line as shown below.



Fix −4 now. See whether 2 is to the right or the left of −4. In this case, 2 is to the right of−4 and in the positive direction. So, 2 > −4 or otherwise −4 < 2.

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